KARYOTYPING DIFFERENT ABNORMAL CASES II

BALANCED STRUCTURAL

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ISCN:

2p25.1

3q13.2

Chromosome Abnormalities

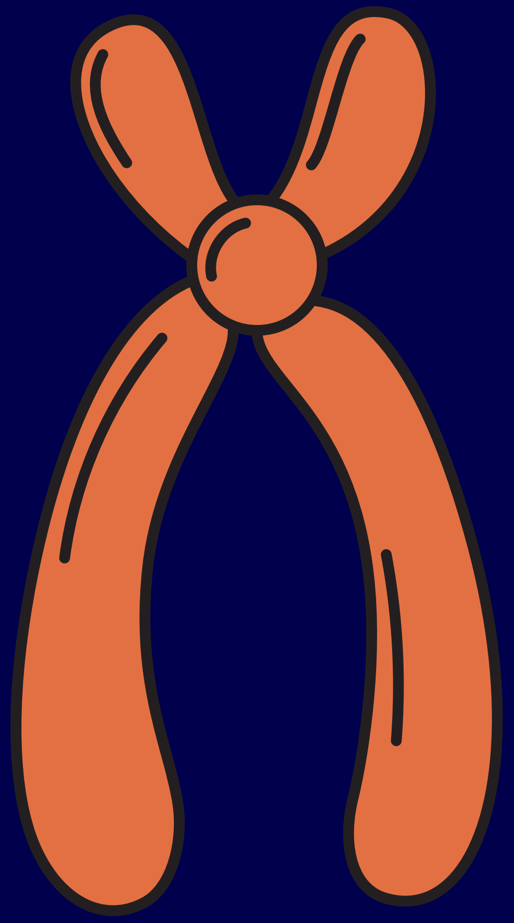
- Numerical
  - Polyploidy
  - Aneuploidy
- Structural
  - Balanced
  - Unbalanced

STRUCTURAL CHROMOSOMAL ABNORMALITIES

- Structural chromosomal abnormalities result from breakage and incorrect rejoining of chromosomal segments.

1. Balanced: without gain or loss of genetic material
2. Unbalanced: gain or loss of genetic material

The indicators used to identify and characterize a structural rearrangement:



- Changes in the G-banding pattern
- The position of the centromere
- The alterations in the short (p) to long (q) arm ratio

The common structural rearrangements in human karyotypes

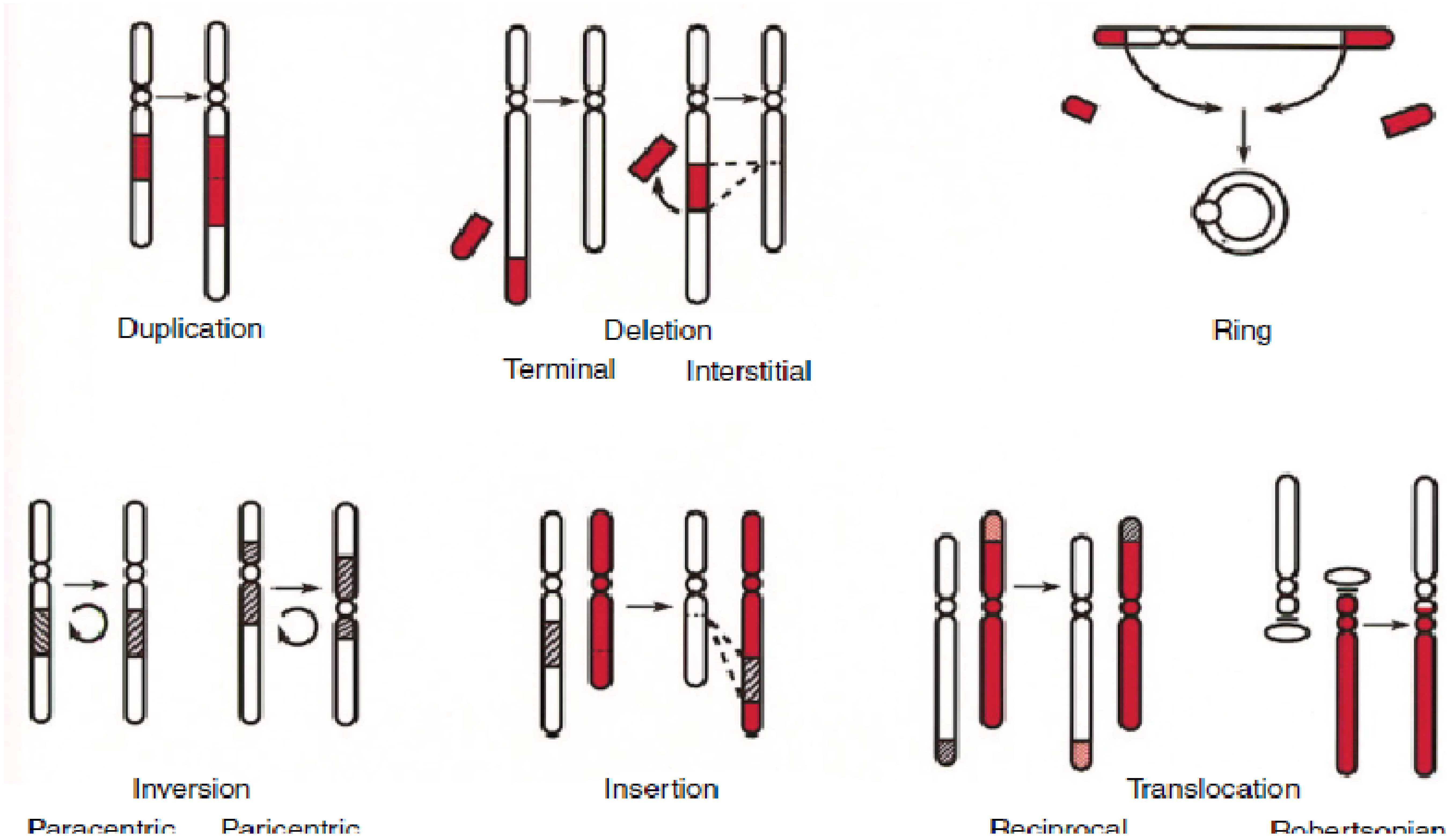


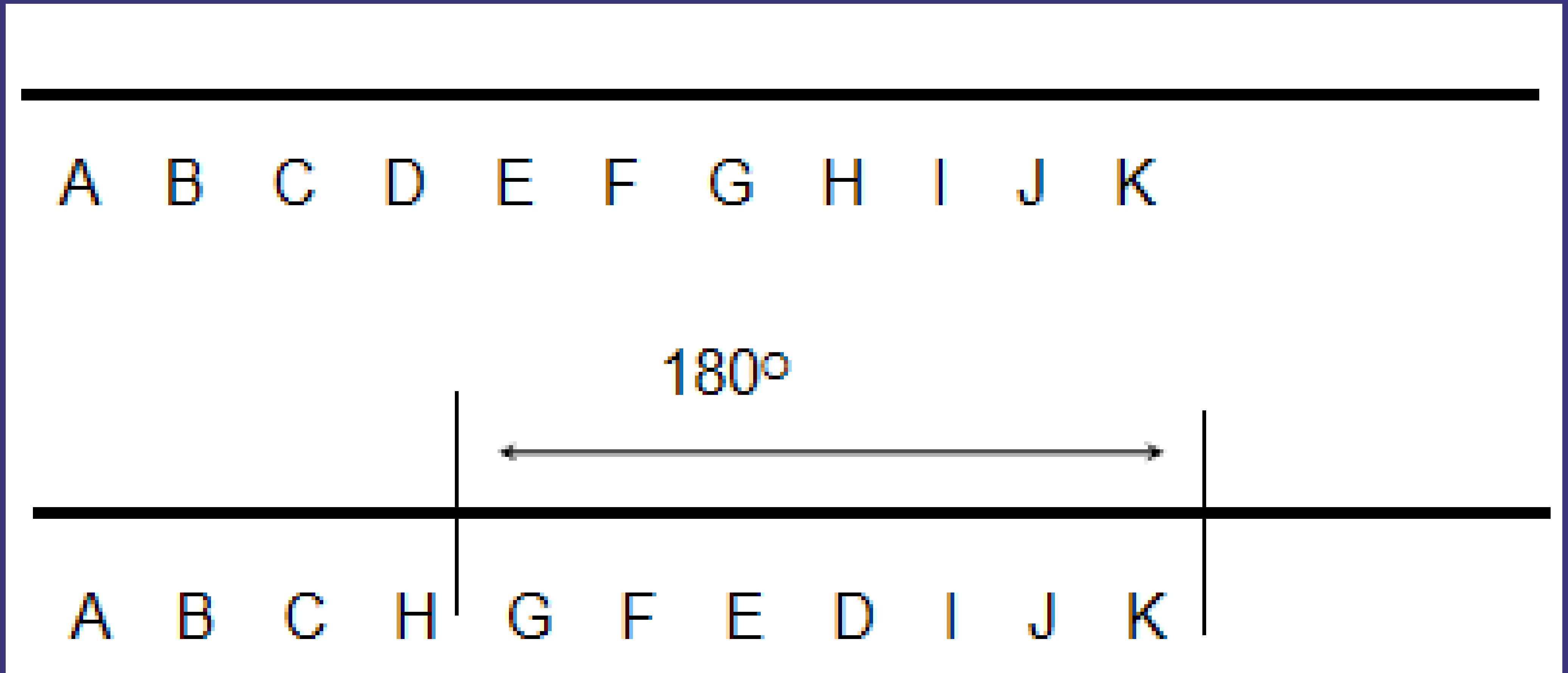
Figure 9.11 The common structural rearrangements observed in human karyotypes include duplications, terminal and interstitial deletions, rings, paracentric and pericentric inversions, insertions and both reciprocal and Robertsonian translocations. See text f

STRUCTURAL CHROMOSOMAL ABNORMALITIES:

- Inversion
- Translocation
- Insertion
- Deletion
- Duplication
- Isochromosome
- Ring chromosome

INVERSION

Inversions: are intrachromosomal rearrangements in which the material located between two breakpoints is rotated 1800 and reinserted in the opposite orientation.



INVERSION

- May change phenotype through "position effects":

1. Lose gene function: move active genes to sites generally inactive.
2. Gain gene function: move inactive genes to sites generally active

INVERSION

- Inversion rearrangements are divided into two groups, depending on the position of the breakpoints:

1. pericentric inversion: is formed by breakpoints on either side of the centromere, one in each Arm
2. paracentric inversion: is produced by breakpoints confined to only one chromosome arm

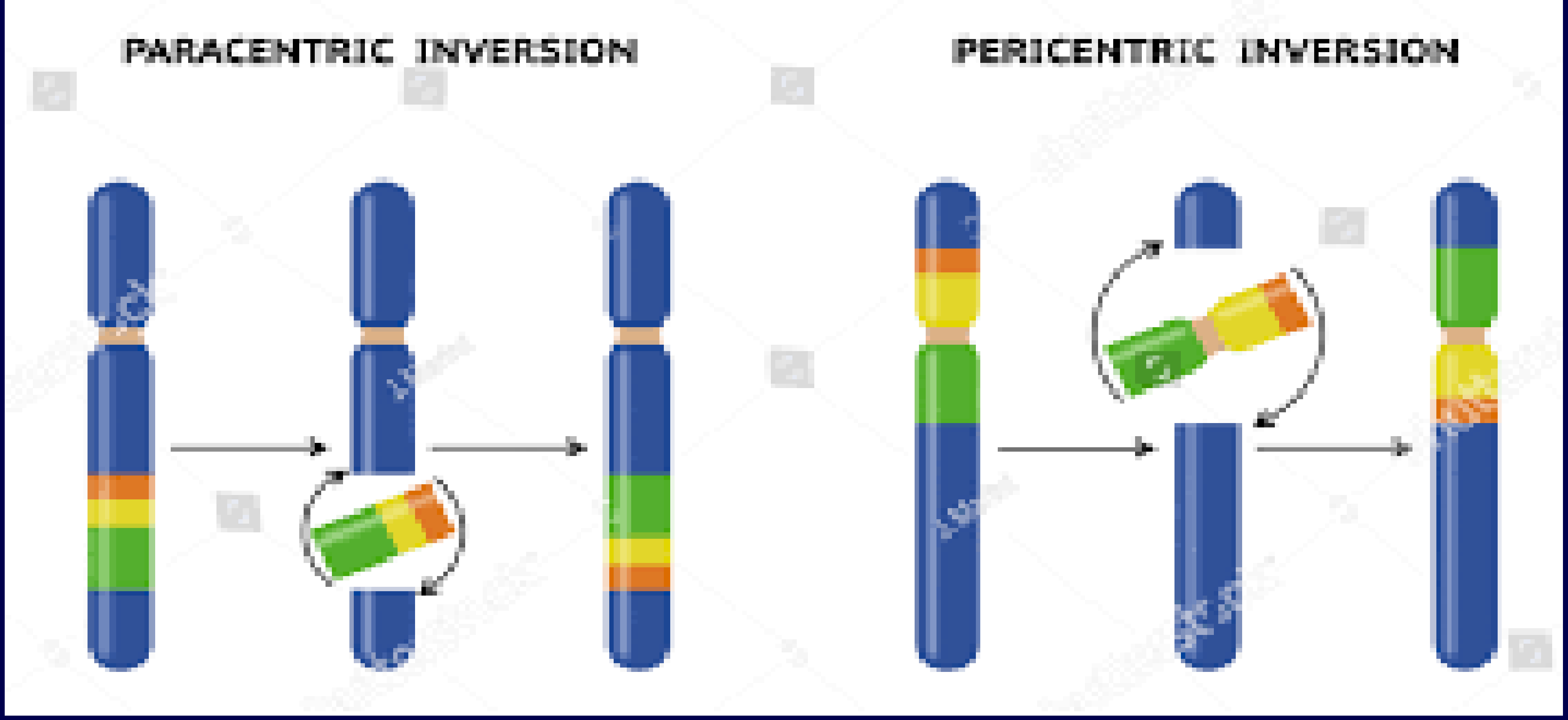
TYPES OF INVERSION

Paracentric inversion:

e.g. 46,XX,inv(9),(q11q32)

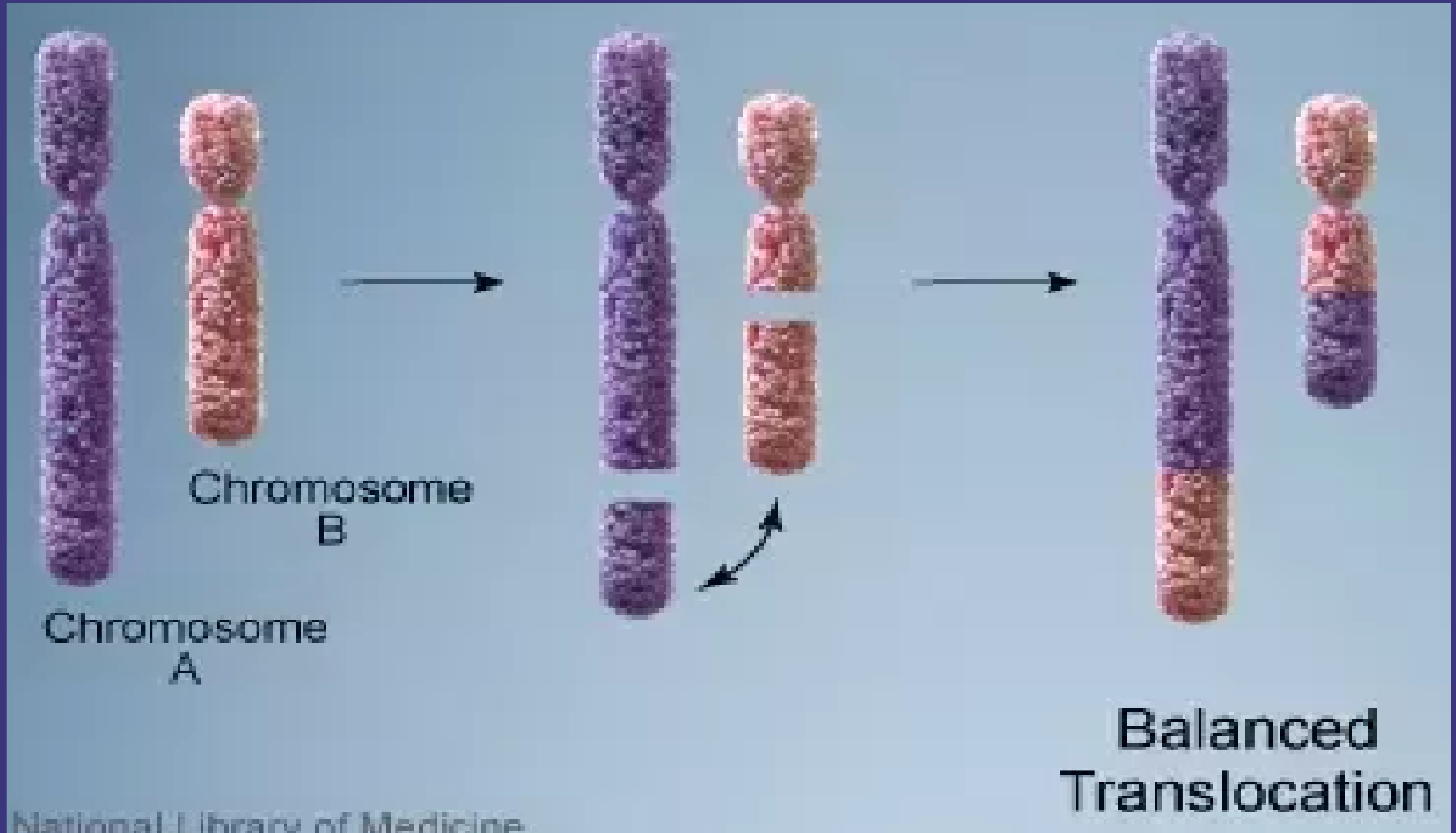
Pericentric inversion:

e.g.46,XX,inv(9),(p11q12)



TRANSLOCATION

Translocation: A genetic change in which a piece of one chromosome breaks off and attaches to another chromosome.



RECIPROCAL TRANSLOCATION

Balanced reciprocal translocation:

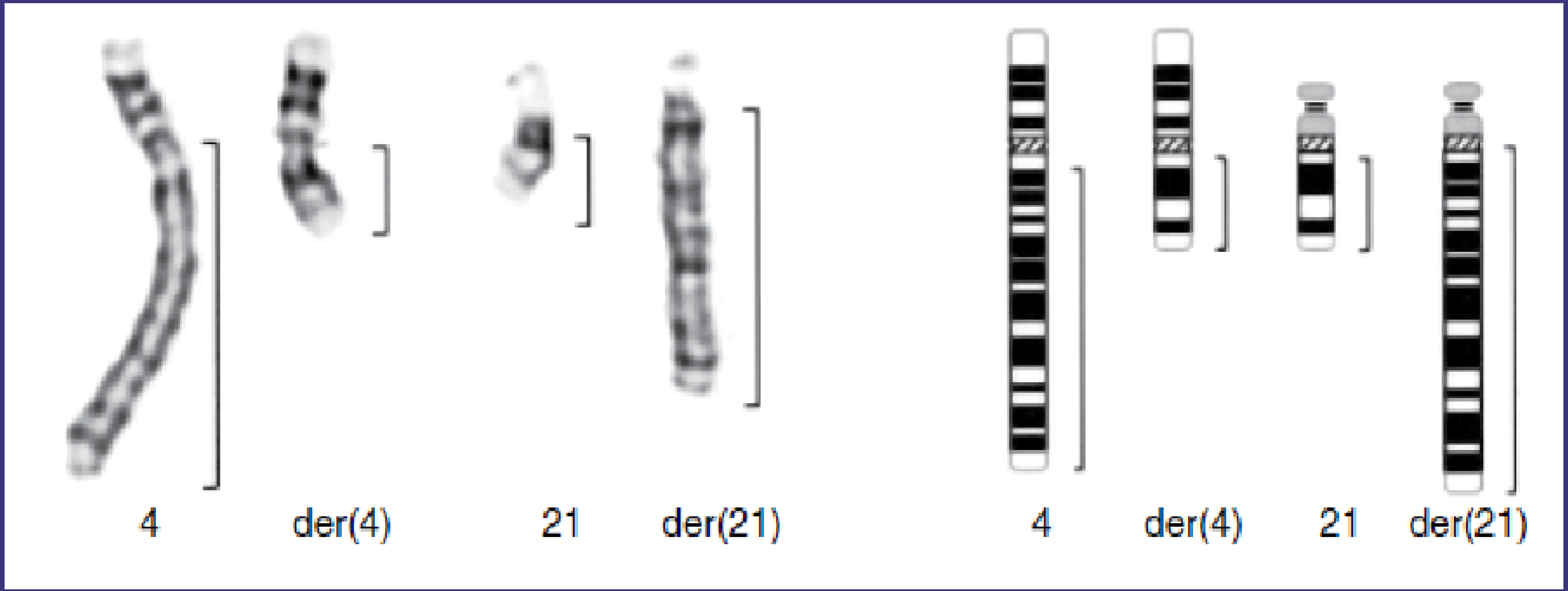
Simple reciprocal translocations result from interchange of parts of two chromosomes without visible loss of chromosomal material

Unbalanced reciprocal translocation:

occur when a fetus inherits a chromosome with extra or missing genetic material from a parent with a balanced translocation

A BALANCED RECIPROCAL TRANSLOCATION INVOLVING THE LONG ARMS OF CHROMOSOMES 4 AND 21.

46,XX,T(4;21)(Q12;Q11.2)



TRANSLOCATION

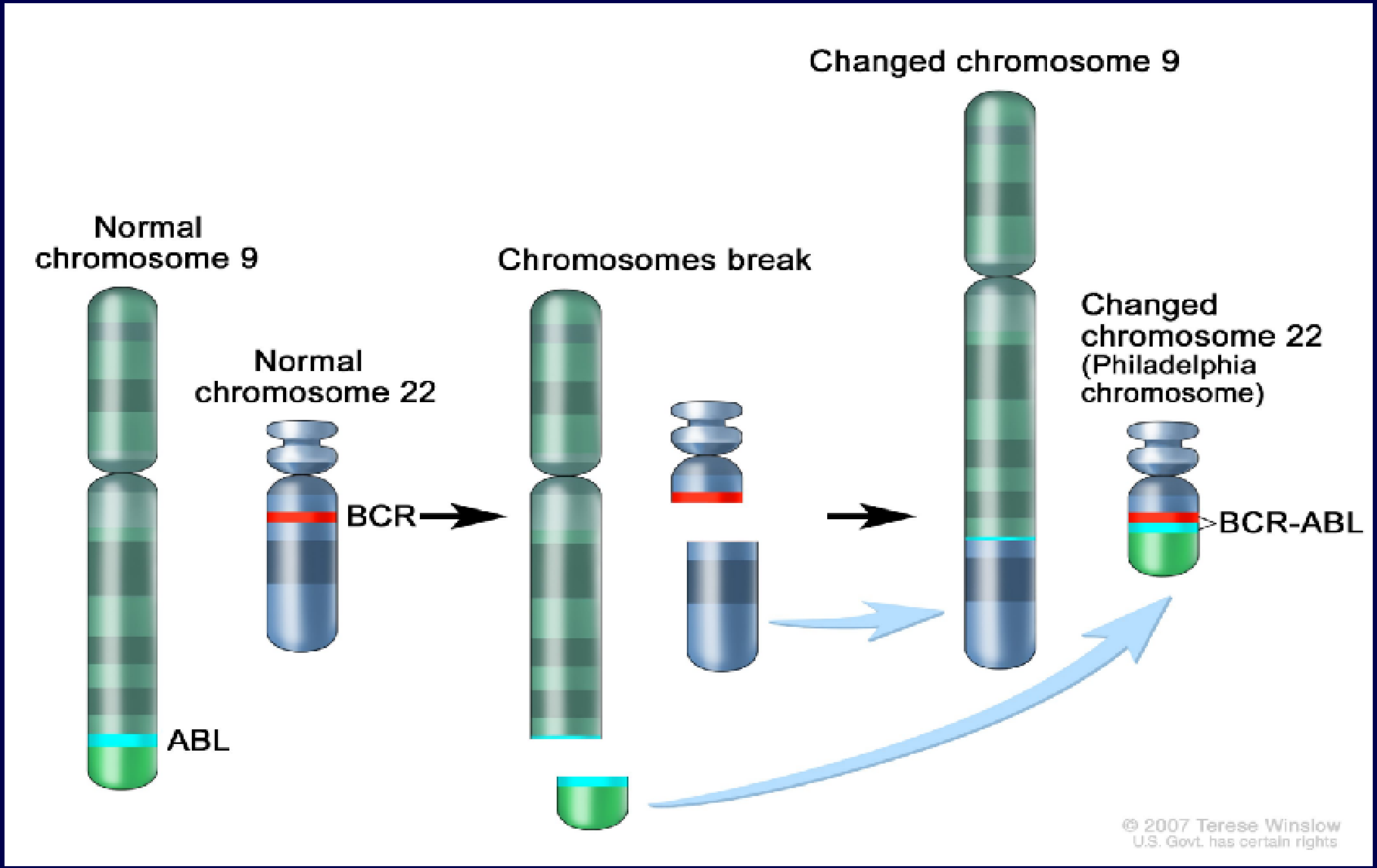
1. Philadelphia translocation

2. Robertsonian translocation

PHILADELPHIA TRANSLOCATION

- Philadelphia translocation: a reciprocal translocation involving chromosomes 9 and 22 that is commonly identified in chronic myelogenous leukemia (CML).
- The break points of the translocation create a fusion of two genes: ABL1 on chromosome 9 and BCR on chromosome 22.

PHILADELPHIA CHROMOSOME



PHILADELPHIA CHROMOSOME

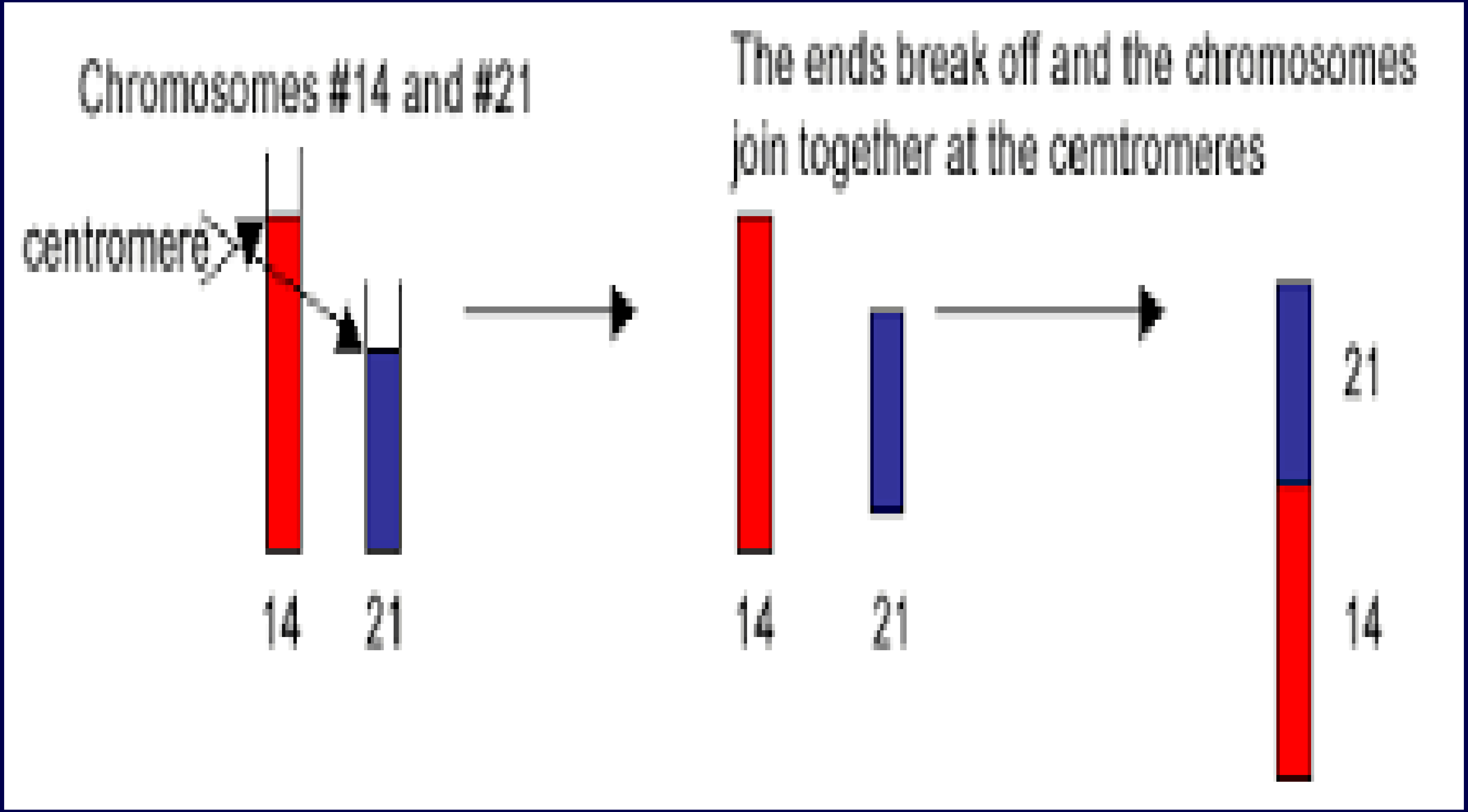
e.g. 46,XX,t(9;22) (q34;q11)

ROBERTSONIAN CHROMOSOME

Acrocentric chromosomes
(13,14,15, 21,22)

Occur when the long arms of any two of the five acrocentric join at or near the centromere to produce one large derivative chromosome.

ROBERTSONIAN CHROMOSOME



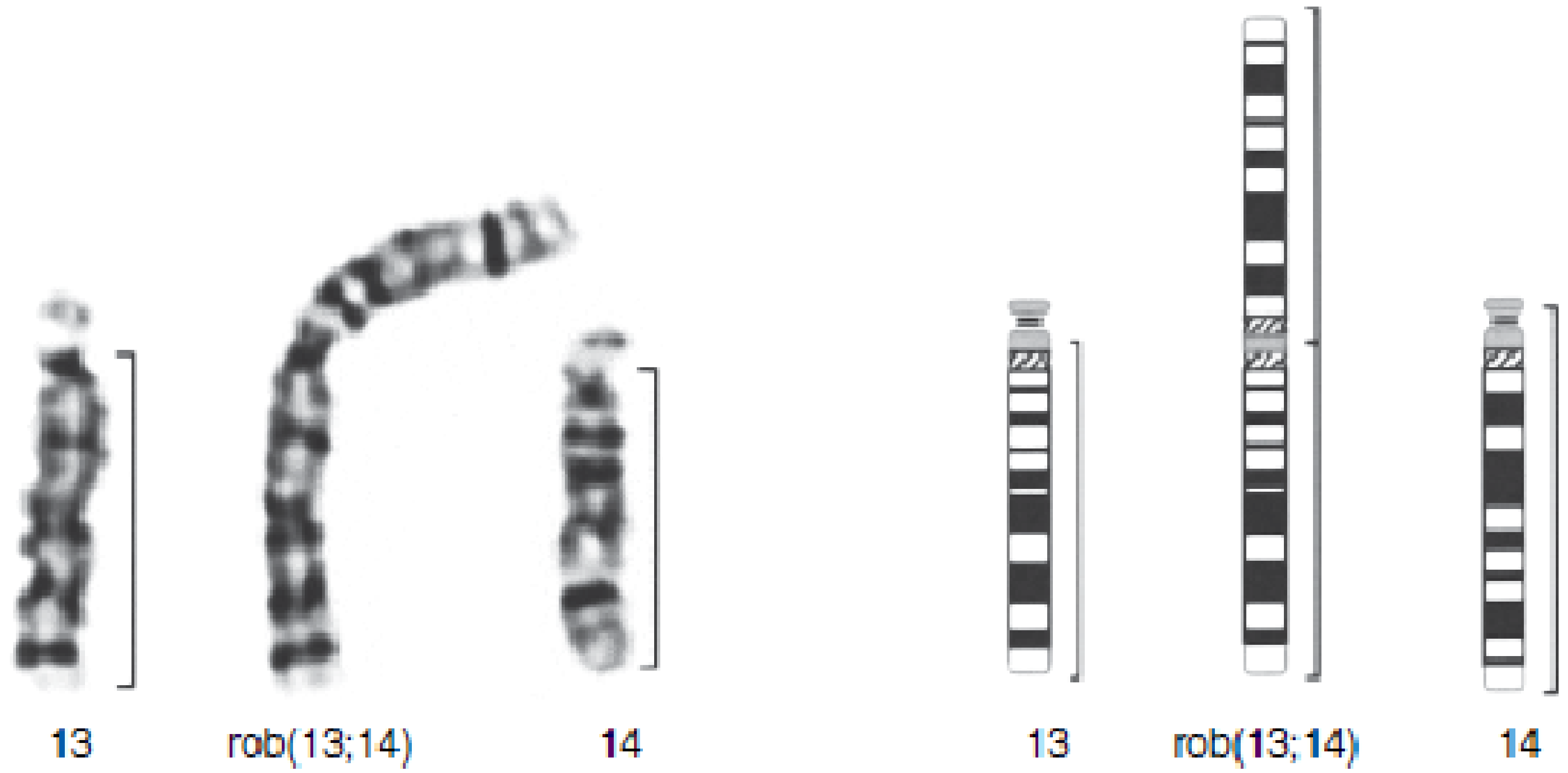


Figure 9.25 A balanced Robertsonian translocation involving chromosomes 13 and 14 [der or rob(13;14)(q10;q10)].

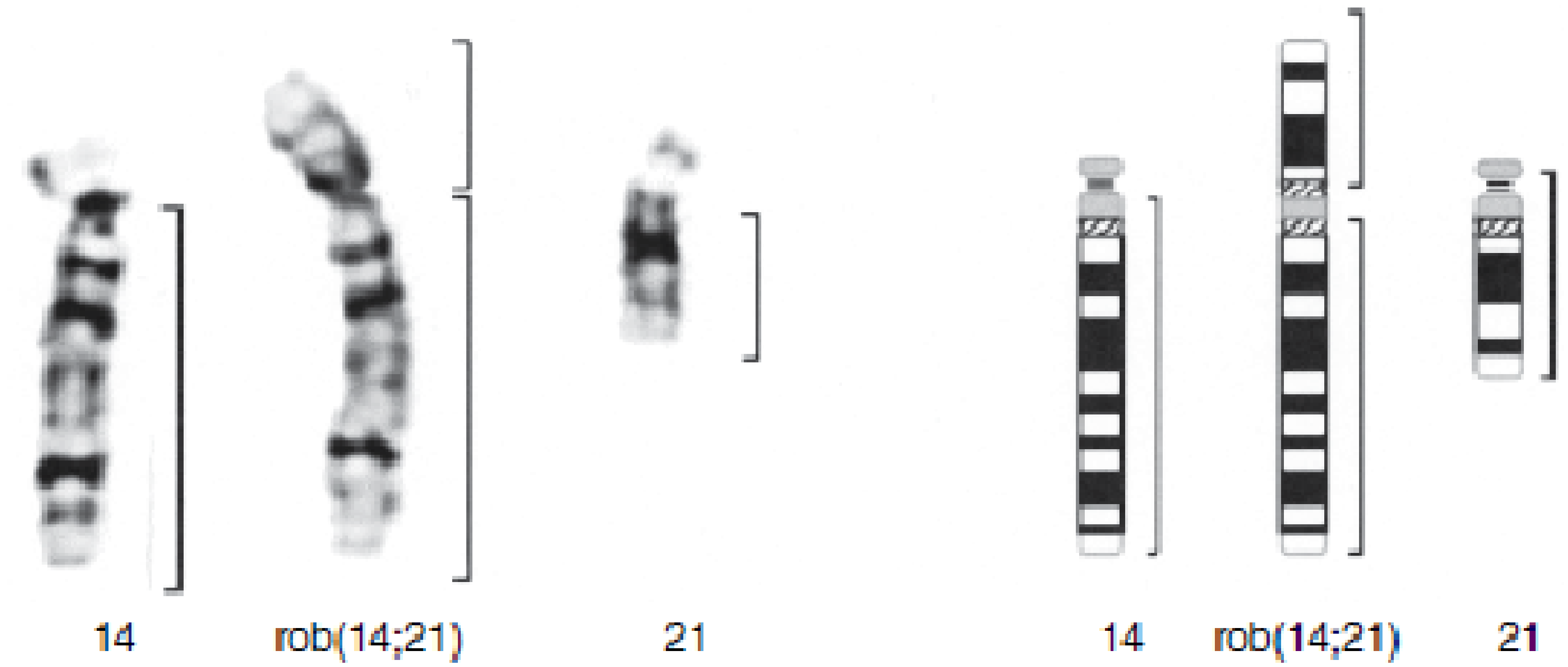
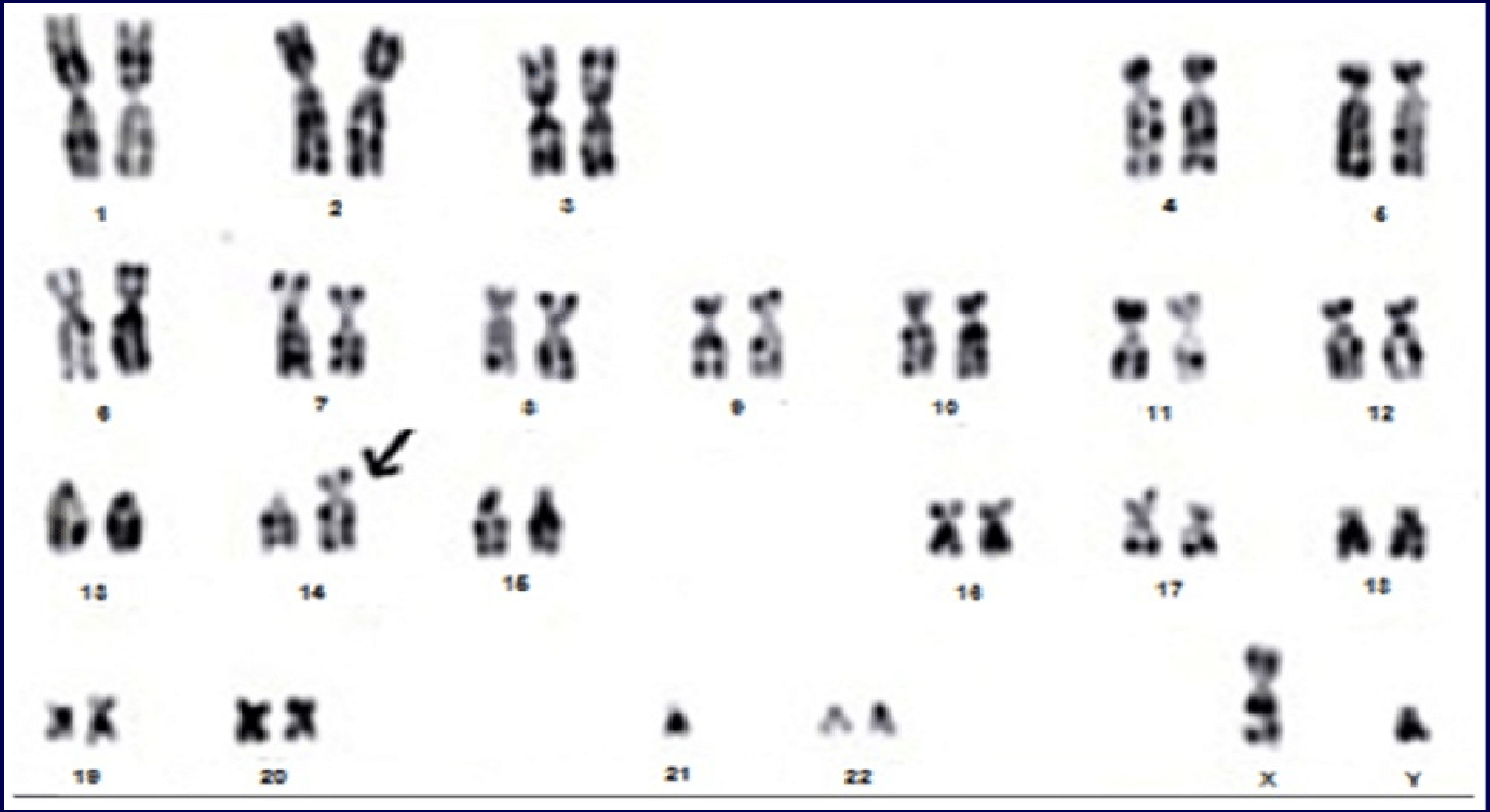


Figure 9.26 A balanced Robertsonian translocation involving chromosomes 14 and 21 [der or rob(14;21)(q10;q10)].



A unique Robertsonian translocation between chromosome 14 and 21 resulting 45,XY,rob(14;21)(q10;q10)

ROBERTSONIAN CHROMOSOME

Most balanced Robertsonian involve different chromosomes

e.g. 45,XX,rob(15;21) (q10;q10)

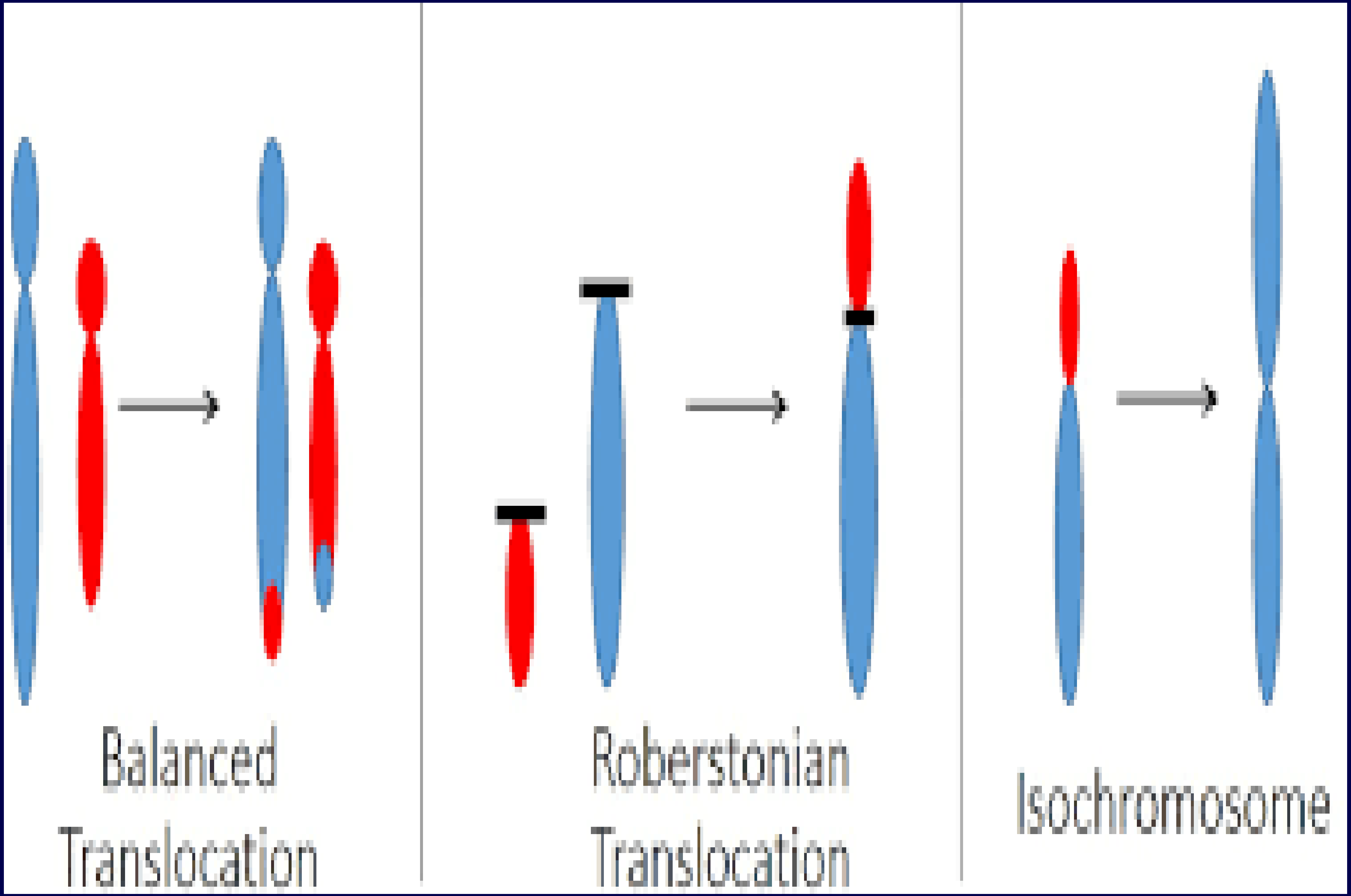
e.g. 45,XX,rob(13;14) (q10;q10)

Fusion of homologous end up with isochromsome

e.g. 45,XY,I (21) (q10)

ROBERTSONIAN CHROMOSOME

- Carrier of a Robertsonian translocation is phenotypically normal but there is a risk for unbalanced gametes and therefore for unbalanced offspring.
- The chief clinical importance of this type of translocation involving chromosome 21 are at risk for producing a child with translocation Down syndrome
- 45,XY,i(21) (q10) parent
- 46,XY,i(21) (q10) child (classical Down syndrome)


Balanced
Translocation
Roberstonian
Translocation
Isochromosome

INSERTION

Insertions: are chromosomal aberrations with a genomic fragment inserted into a nonhomologous chromosome (interchromosomal rearrangements) and or into a different locus (the same or other arm) on the same chromosome or the other homolog (intrachromosomal rearrangements)

INSERTION

Insertion within a Chromosome

46,XX,ins(3) (p21q27q32)

This represents a direct insertion within a chromosome ,the long arm segments between band 3q27 and 3q32 have been inserted into the short arm of the same chromosome at band 3p21.

INSERTION

- Insertion Involving Two Chromosomes
- 46,XX,ins(4;9) (q31;q12q13)
- The long arm segments between bands 9q12 and 9q13 have been inserted into the long arm of chromosome 4 at band q31.

- The recipient chromosome is always specified first regardless of the chromosome number.

NOTES

Question: If a person has both numerical +21 and structural translocation t(3;5) (q12;q56) how do we write it?

Answer: So, start with numerical then structural.

NOTES

Question: If a translocation happened between an autosome and sex chromosome how do we write this case?

Answer: So, start with sex then autosome chromosome